

# Assignment\_06CT

September 18, 2026

## 1 Kushagra Singh 2401PH07 Assignment 6

### 2 Q1

```
[44]: import math
import matplotlib.pyplot as plt
## Implementing Cot(x)

# Given data
x = [1, 2, 3, 4, 5] # degrees
f = [57.28996163, 28.63625328, 19.08113669,
     14.30066626, 11.43005230]
# x=1'
h=1
#x=float(input("Enter x Value for diffrentiation"))
## Logic Forward Difference
forward=(f[1]-f[0])/h
# x=2'
## Logic Central Difference
central=(f[2]-f[0])/(h*2)
print("forward diffrence approach :", forward)
print("centraldiffrence approach :", central)

F=[]
C=[]
## Using Loops
for i in range(4):
    forward=(f[i+1]-f[i])/h
    F.append(forward)
for j in range(1,4):
    central=(f[j+1]-f[j-1])/(2*h)
    C.append(central)
print("Forward", F)
print("Central", C)
x_forward=x[:-1]
x_central=x[1:-1]
plt.plot(x,f,label="Function")
```

```

plt.plot(x_central,C,'^--',label="Central Difference")
plt.plot(x_forward,F,label="Forward Differnce")
plt.xlabel("X values")
plt.ylabel("Fuction value Cot(x)")
plt.title("Cot (x), Graph")
plt.legend()
plt.show()

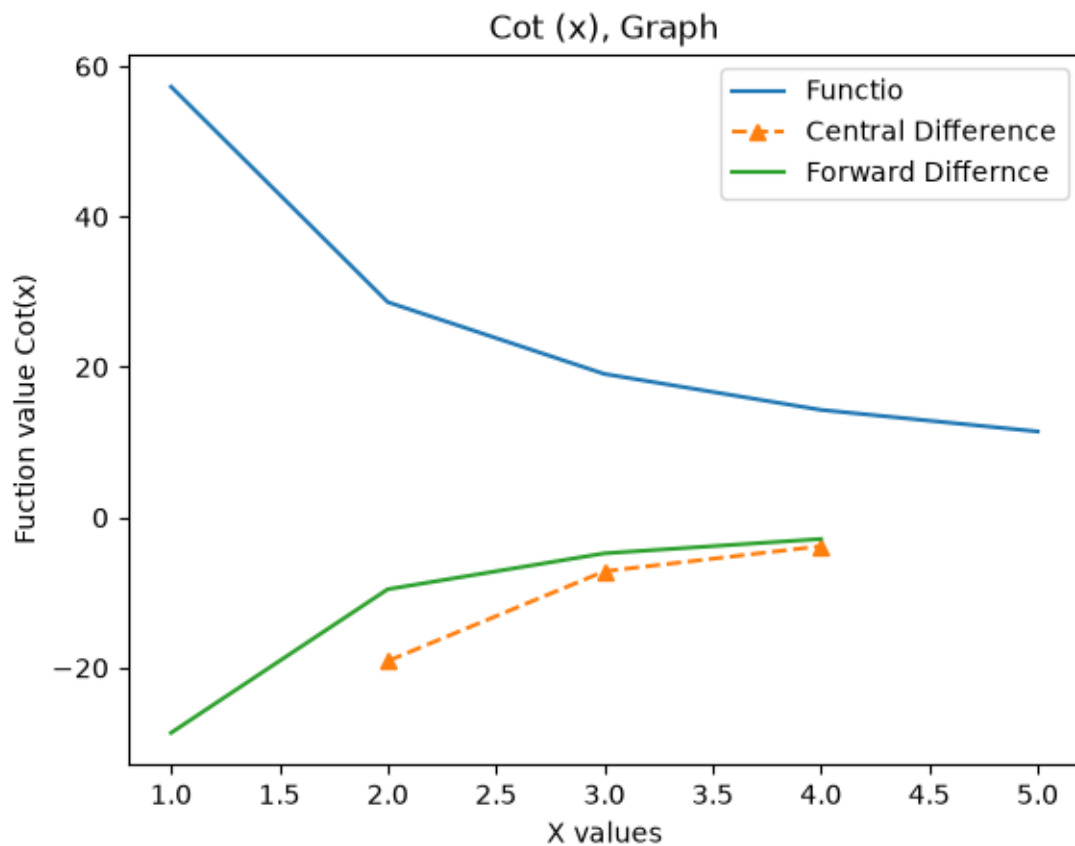
```

forward differnce approach : -28.653708350000002

centraldifference approach : -19.10441247

Forward [-28.653708350000002, -9.555116589999997, -4.780470430000001, -2.87061396]

Central [-19.10441247, -7.167793509999999, -3.8255421950000006]



### 3 Q2

```
[58]: t = [0.0, 0.1, 0.2, 0.3, 0.4] # seconds
h=0.1
y = [1.200, 1.150, 1.010, 0.780, 0.460] # meters
C=(y[3]-y[1])/(2*h)
print("Value at 0.2s",C,"m/s")
A=(y[4]-y[2])/(2*h)
print("Value at 0.3s",A,"m/s")

### actual values v=a*t
a=9.8
v1=-9.8*t[2]
v2=-9.8*t[3]
print("Actual value at 0.2s:",v1,"m/s")
print("Actual value at 0.3s:",v2,"m/s")
error1=C-v1
error2=A-v2
print("Error in velocity at 0.2s", error1)
print("Error in velocity at 0.3s", error2)
```

```
Value at 0.2s -1.8499999999999994 m/s
Value at 0.3s -2.75 m/s
Actual value at 0.2s: -1.9600000000000002 m/s
Actual value at 0.3s: -2.94 m/s
Error in velocity at 0.2s 0.110000000000000076
Error in velocity at 0.3s 0.18999999999999995
```

### 4 Q3

```
[61]: import math
def f(x):
    return math.exp(-x) * math.sin(x)

x = 1.0
h = 0.1
D_h = (f(x + h) - f(x - h)) / (2 * h)
D_h2 = (f(x + h/2) - f(x - h/2)) / (2 * (h/2))
D_R = (4 * D_h2 - D_h) / 3
print("Central Difference D(h) :", D_h)
print("Central Difference D(h/2) :", D_h2)
print("Richardson Extrapolation :", D_R)
exact = math.exp(-x) * (math.cos(x) - math.sin(x))
print("Exact derivative:", exact)
error = abs(D_R - exact)
print("Error:", error)
```

Central Difference  $D(h)$  : -0.10909897684773184  
Central Difference  $D(h/2)$  : -0.11037013724893963  
Richardson Extrapolation : -0.11079385738267555  
Exact derivative: -0.11079376530669924  
Error: 9.207597631566422e-08

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